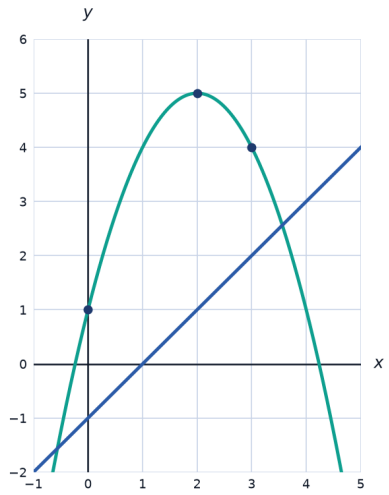
# Negative exponents

## Core Concepts: Negative exponents

**1** For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(2)$.




**a.** 11

**b.** 15

**c.** 8

**d.** 15

**2** Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

**a.** $x = 1$ and $x = 6$

**b.** $x = 1$ and $x = 5$

**c.** $x = -1$ and $x = -5$

**d.** $x = 6$ and $x = 5$

**3** A data set recorded for negative exponents consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5

---

## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(3)$.

**a.** 15

**b.** 19

**c.** 12

**d.** 19

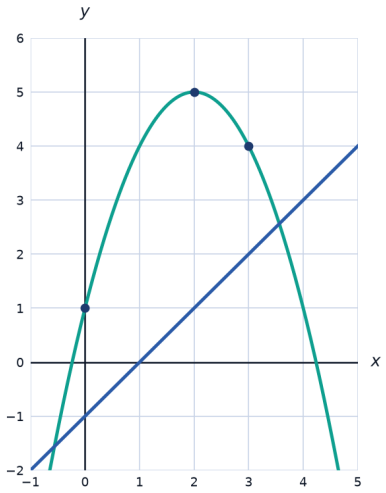
**5** Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

**a.** $x = 2$ and $x = 7$

**b.** $x = 2$ and $x = 6$

**c.** $x = -2$ and $x = -6$

**d.** $x = 8$ and $x = 12$




**6** A data set recorded for negative exponents consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5

---

## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(4)$.

**a.** 19

**b.** 23

**c.** 16

**d.** 23

**8** Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

**a.** $x = 3$ and $x = 8$

**b.** $x = 3$ and $x = 7$

**c.** $x = -3$ and $x = -7$

**d.** $x = 10$ and $x = 21$

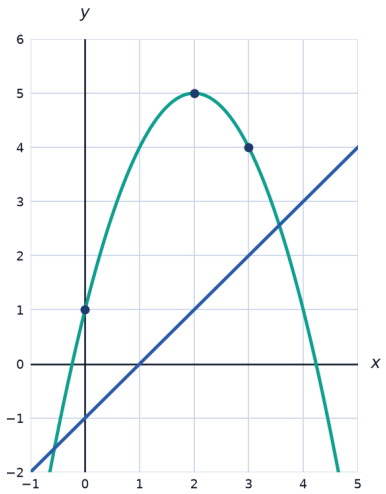
**9** A data set recorded for negative exponents consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

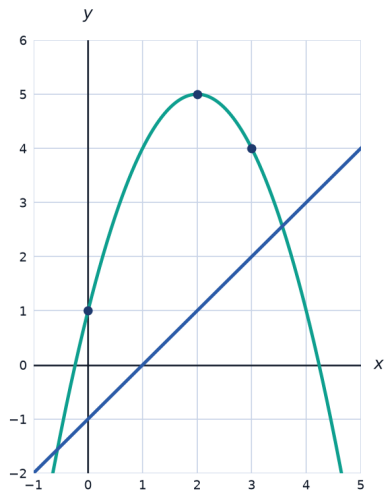
**d.** 24.5




## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(5)$.




**a.** 23

**b.** 27

**c.** 20

**d.** 27

**11** Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

**a.** $x = 4$ and $x = 9$

**b.** $x = 4$ and $x = 8$

**c.** $x = -4$ and $x = -8$

**d.** $x = 12$ and $x = 32$

**12** A data set recorded for negative exponents consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5

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